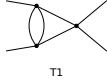


SK Matrix-Model Contributions for All L=2 Topologies

External assignment (by external-propagator appearance): $\phi_r^1(x_1) \phi_a^1(x_2) \phi_r^2(x_3) \phi_a^2(x_4)$, with $x_1 = (t, \mathbf{x})$, $x_2 = (0, \mathbf{0})$, $x_3 = (t, \mathbf{x})$, $x_4 = (0, \mathbf{0})$. Translation table used: same-fold $G_{rr} \rightarrow G_K = \frac{1}{2}(G^> + G^<)$, $G_{ra} \rightarrow G_R$, $G_{ar} \rightarrow G_A$, $G_{aa} \rightarrow 0$; cross-fold $G_{rr}^{12} \rightarrow G^<$, $G_{rr}^{21} \rightarrow G^>$, and any cross-fold propagator with an a endpoint vanishes. Contour ordering convention: contour (2) is later than contour (1), matching `contour_ra_translation.txt`. Each term carries explicit net matrix-size factor N_c^{p-4} (raw loop count p minus the overall $1/N^4$ normalization).

Topology 1 (FeynArts Topology 2)

2 → 2



τ1

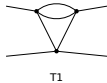
Parsed propagators: 8. Internal vertices: 3. Internal half-edges: 12.
Assignments checked: 32768. Surviving (nonzero vertex weight): 4096.
Raw N_c power: 0. Net power in $C(t)$: -4. Explicit surviving terms:

$$\mathcal{T}_{1,1} = 128ig_2^3 N_c^{-4} G^<(z_1, z_3) G^<(z_2, z_3) G_A(x_2, z_2) G_A(x_4, z_3) G_A(z_2, z_1) G_K(x_1, z_1) G_R(x_3, z_3) G_R(z_2, z_1)$$

$$\mathcal{T}_{1,2} = 128ig_2^3 N_c^{-4} G^<(z_1, z_3) G^<(z_2, z_3) G_A(x_2, z_2) G_A(x_4, z_3) G_A(z_2, z_1) G_K(z_2, z_1) G_R(x_1, z_1) G_R(x_3, z_3)$$

Topology 2 (FeynArts Topology 2)

2 → 2



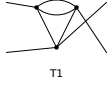
τ1

Parsed propagators: 8. Internal vertices: 3. Internal half-edges: 12.
Assignments checked: 32768. Surviving (nonzero vertex weight): 4096.
Raw N_c power: 0. Net power in $C(t)$: -4.

No surviving SK contributions for this topology.

Topology 3 (FeynArts Topology 2)

$2 \rightarrow 2$

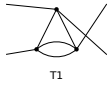


Parsed propagators: 8. Internal vertices: 3. Internal half-edges: 12.
 Assignments checked: 32768. Surviving (nonzero vertex weight): 4096.
 Raw N_c power: 0. Net power in $C(t)$: -4.

No surviving SK contributions for this topology.

Topology 4 (FeynArts Topology 2)

$2 \rightarrow 2$

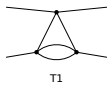


Parsed propagators: 8. Internal vertices: 3. Internal half-edges: 12.
 Assignments checked: 32768. Surviving (nonzero vertex weight): 4096.
 Raw N_c power: 0. Net power in $C(t)$: -4.

No surviving SK contributions for this topology.

Topology 5 (FeynArts Topology 2)

$2 \rightarrow 2$

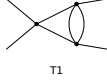


Parsed propagators: 8. Internal vertices: 3. Internal half-edges: 12.
Assignments checked: 32768. Surviving (nonzero vertex weight): 4096.
Raw N_c power: 0. Net power in $C(t)$: -4.

No surviving SK contributions for this topology.

Topology 6 (FeynArts Topology 2)

2 → 2



τ₁

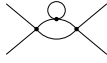
Parsed propagators: 8. Internal vertices: 3. Internal half-edges: 12.
Assignments checked: 32768. Surviving (nonzero vertex weight): 4096.
Raw N_c power: 0. Net power in $C(t)$: -4. Explicit surviving terms:

$$\mathcal{T}_{6,1} = 128ig_2^3 N_c^{-4} G^>(z_2, z_1) G^>(z_3, z_1) G_A(x_2, z_1) G_A(x_4, z_3) G_A(z_3, z_2) G_K(x_3, z_2) G_R(x_1, z_1) G_R(z_3, z_2)$$

$$\mathcal{T}_{6,2} = 128ig_2^3 N_c^{-4} G^>(z_2, z_1) G^>(z_3, z_1) G_A(x_2, z_1) G_A(x_4, z_3) G_A(z_3, z_2) G_K(z_3, z_2) G_R(x_1, z_1) G_R(x_3, z_2)$$

Topology 7 (FeynArts Topology 2)

2 → 2



τ₁

Parsed propagators: 8. Internal vertices: 3. Internal half-edges: 12.
Assignments checked: 32768. Surviving (nonzero vertex weight): 4096.
Raw N_c power: 1. Net power in $C(t)$: -3. Explicit surviving terms:

$$\mathcal{T}_{7,1} = 64ig_2^3 N_c^{-3} G^<(z_1, z_2) G^<(z_1, z_3) G_A(x_2, z_1) G_A(x_4, z_2) G_A(z_2, z_3) G_A(z_3, z_3) G_K(x_3, z_2) G_R(x_1, z_1)$$

$$\mathcal{T}_{7,2} = 64ig_2^3 N_c^{-3} G^<(z_1, z_2) G^<(z_1, z_3) G_A(x_2, z_1) G_A(x_4, z_2) G_A(z_2, z_3) G_K(x_3, z_2) G_R(x_1, z_1) G_R(z_3, z_3)$$

$$\mathcal{T}_{7,3} = 64ig_2^3 N_c^{-3} G^<(z_1, z_2) G^<(z_1, z_3) G_A(x_2, z_1) G_A(x_4, z_2) G_A(z_3, z_3) G_K(z_2, z_3) G_R(x_1, z_1) G_R(x_3, z_2)$$

$$\mathcal{T}_{7,4} = 64ig_2^3 N_c^{-3} G^<(z_1, z_2) G^<(z_1, z_3) G_A(x_2, z_1) G_A(x_4, z_2) G_K(z_2, z_3) G_R(x_1, z_1) G_R(x_3, z_2) G_R(z_3, z_3)$$

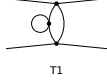
$$\mathcal{T}_{7,5} = 64ig_2^3 N_c^{-3} G^<(z_1, z_2) G^<(z_1, z_3) G_A(x_2, z_1) G_A(x_4, z_2) G_K(z_3, z_3) G_R(x_1, z_1) G_R(x_3, z_2) G_R(z_2, z_3)$$

$$\mathcal{T}_{7,6} = 64ig_2^3 N_c^{-3} G^<(z_1, z_2) G^>(z_2, z_3) G_A(x_2, z_1) G_A(x_4, z_2) G_A(z_1, z_3) G_A(z_3, z_3) G_K(x_1, z_1) G_R(x_3, z_2)$$

$$\begin{aligned}
\mathcal{T}_{7,7} &= 64ig_2^3 N_c^{-3} G^<(z_1, z_2) G^>(z_2, z_3) G_A(x_2, z_1) G_A(x_4, z_2) G_A(z_1, z_3) G_K(x_1, z_1) G_R(x_3, z_2) G_R(z_3, z_3) \\
\mathcal{T}_{7,8} &= 64ig_2^3 N_c^{-3} G^<(z_1, z_2) G^>(z_2, z_3) G_A(x_2, z_1) G_A(x_4, z_2) G_A(z_3, z_3) G_K(z_1, z_3) G_R(x_1, z_1) G_R(x_3, z_2) \\
\mathcal{T}_{7,9} &= 64ig_2^3 N_c^{-3} G^<(z_1, z_2) G^>(z_2, z_3) G_A(x_2, z_1) G_A(x_4, z_2) G_K(z_1, z_3) G_R(x_1, z_1) G_R(x_3, z_2) G_R(z_3, z_3) \\
\mathcal{T}_{7,10} &= 64ig_2^3 N_c^{-3} G^<(z_1, z_2) G^>(z_2, z_3) G_A(x_2, z_1) G_A(x_4, z_2) G_K(z_3, z_3) G_R(x_1, z_1) G_R(x_3, z_2) G_R(z_1, z_3)
\end{aligned}$$

Topology 8 (FeynArts Topology 2)

2 → 2



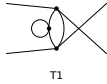
T1

Parsed propagators: 8. Internal vertices: 3. Internal half-edges: 12.
Assignments checked: 32768. Surviving (nonzero vertex weight): 4096.
Raw N_c power: 1. Net power in $C(t)$: -3.

No surviving SK contributions for this topology.

Topology 9 (FeynArts Topology 2)

2 → 2



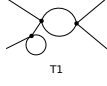
T1

Parsed propagators: 8. Internal vertices: 3. Internal half-edges: 12.
Assignments checked: 32768. Surviving (nonzero vertex weight): 4096.
Raw N_c power: 1. Net power in $C(t)$: -3.

No surviving SK contributions for this topology.

Topology 10 (FeynArts Topology 4)

2 → 2



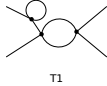
T1

Parsed propagators: 8. Internal vertices: 3. Internal half-edges: 12.
 Assignments checked: 32768. Surviving (nonzero vertex weight): 4096.
 Raw N_c power: 1. Net power in $C(t)$: -3. Explicit surviving terms:

$$\begin{aligned}
 \mathcal{T}_{10,1} &= 128ig_2^3 N_c^{-3} G^<(x_1, z_1) G^<(z_2, z_1) G_A(x_2, z_2) G_A(x_4, z_3) G_A(z_1, z_3) G_A(z_2, z_2) G_K(x_3, z_3) G_R(z_1, z_3) \\
 \mathcal{T}_{10,2} &= 128ig_2^3 N_c^{-3} G^<(x_1, z_1) G^<(z_2, z_1) G_A(x_2, z_2) G_A(x_4, z_3) G_A(z_1, z_3) G_A(z_2, z_2) G_K(z_1, z_3) G_R(x_3, z_3) \\
 \mathcal{T}_{10,3} &= 128ig_2^3 N_c^{-3} G^<(x_1, z_1) G^<(z_2, z_1) G_A(x_2, z_2) G_A(x_4, z_3) G_A(z_1, z_3) G_K(x_3, z_3) G_R(z_1, z_3) G_R(z_2, z_2) \\
 \mathcal{T}_{10,4} &= 128ig_2^3 N_c^{-3} G^<(x_1, z_1) G^<(z_2, z_1) G_A(x_2, z_2) G_A(x_4, z_3) G_A(z_1, z_3) G_K(z_1, z_3) G_R(x_3, z_3) G_R(z_2, z_2) \\
 \mathcal{T}_{10,5} &= 64ig_2^3 N_c^{-3} G^<(z_1, z_3) G^<(z_1, z_3) G_A(x_2, z_2) G_A(x_4, z_3) G_A(z_2, z_1) G_K(z_2, z_2) G_R(x_1, z_1) G_R(x_3, z_3) \\
 \mathcal{T}_{10,6} &= 64ig_2^3 N_c^{-3} G^<(z_1, z_3) G^<(z_1, z_3) G_A(x_2, z_2) G_A(x_4, z_3) G_A(z_2, z_2) G_K(x_1, z_1) G_R(x_3, z_3) G_R(z_2, z_1) \\
 \mathcal{T}_{10,7} &= 64ig_2^3 N_c^{-3} G^<(z_1, z_3) G^<(z_1, z_3) G_A(x_2, z_2) G_A(x_4, z_3) G_A(z_2, z_2) G_K(z_2, z_1) G_R(x_1, z_1) G_R(x_3, z_3) \\
 \mathcal{T}_{10,8} &= 64ig_2^3 N_c^{-3} G^<(z_1, z_3) G^<(z_1, z_3) G_A(x_2, z_2) G_A(x_4, z_3) G_K(x_1, z_1) G_R(x_3, z_3) G_R(z_2, z_1) G_R(z_2, z_2) \\
 \mathcal{T}_{10,9} &= 64ig_2^3 N_c^{-3} G^<(z_1, z_3) G^<(z_1, z_3) G_A(x_2, z_2) G_A(x_4, z_3) G_K(z_2, z_1) G_R(x_1, z_1) G_R(x_3, z_3) G_R(z_2, z_2)
 \end{aligned}$$

Topology 11 (FeynArts Topology 4)

2 → 2



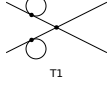
T1

Parsed propagators: 8. Internal vertices: 3. Internal half-edges: 12.
 Assignments checked: 32768. Surviving (nonzero vertex weight): 4096.
 Raw N_c power: 1. Net power in $C(t)$: -3. Explicit surviving terms:

$$\begin{aligned}
 \mathcal{T}_{11,1} &= 64ig_2^3 N_c^{-3} G^<(z_2, z_3) G^<(z_2, z_3) G_A(x_2, z_2) G_A(x_4, z_3) G_A(z_1, z_1) G_A(z_2, z_1) G_K(x_1, z_1) G_R(x_3, z_3) \\
 \mathcal{T}_{11,2} &= 64ig_2^3 N_c^{-3} G^<(z_2, z_3) G^<(z_2, z_3) G_A(x_2, z_2) G_A(x_4, z_3) G_A(z_2, z_1) G_K(x_1, z_1) G_R(x_3, z_3) G_R(z_1, z_1) \\
 \mathcal{T}_{11,3} &= 64ig_2^3 N_c^{-3} G^<(z_2, z_3) G^<(z_2, z_3) G_A(x_2, z_2) G_A(x_4, z_3) G_A(z_2, z_1) G_K(z_1, z_1) G_R(x_1, z_1) G_R(x_3, z_3)
 \end{aligned}$$

Topology 12 (FeynArts Topology 4)

2 → 2

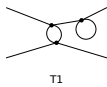


Parsed propagators: 8. Internal vertices: 3. Internal half-edges: 12.
 Assignments checked: 32768. Surviving (nonzero vertex weight): 4096.
 Raw N_c power: 2. Net power in $C(t)$: -2. Explicit surviving terms:

$$\begin{aligned}
 \mathcal{T}_{12,1} &= 64ig_2^3 N_c^{-2} G^<(x_1, z_1) G^<(z_2, z_3) G_A(x_2, z_2) G_A(x_4, z_3) G_A(z_1, z_1) G_A(z_2, z_2) G_K(x_3, z_3) G_R(z_1, z_3) \\
 \mathcal{T}_{12,2} &= 64ig_2^3 N_c^{-2} G^<(x_1, z_1) G^<(z_2, z_3) G_A(x_2, z_2) G_A(x_4, z_3) G_A(z_1, z_1) G_A(z_2, z_2) G_K(z_1, z_3) G_R(x_3, z_3) \\
 \mathcal{T}_{12,3} &= 64ig_2^3 N_c^{-2} G^<(x_1, z_1) G^<(z_2, z_3) G_A(x_2, z_2) G_A(x_4, z_3) G_A(z_1, z_1) G_K(x_3, z_3) G_R(z_1, z_3) G_R(z_2, z_2) \\
 \mathcal{T}_{12,4} &= 64ig_2^3 N_c^{-2} G^<(x_1, z_1) G^<(z_2, z_3) G_A(x_2, z_2) G_A(x_4, z_3) G_A(z_1, z_1) G_K(z_1, z_3) G_R(x_3, z_3) G_R(z_2, z_2) \\
 \mathcal{T}_{12,5} &= 64ig_2^3 N_c^{-2} G^<(x_1, z_1) G^<(z_2, z_3) G_A(x_2, z_2) G_A(x_4, z_3) G_A(z_1, z_3) G_A(z_2, z_2) G_K(z_1, z_1) G_R(x_3, z_3) \\
 \mathcal{T}_{12,6} &= 64ig_2^3 N_c^{-2} G^<(x_1, z_1) G^<(z_2, z_3) G_A(x_2, z_2) G_A(x_4, z_3) G_A(z_1, z_3) G_K(z_1, z_1) G_R(x_3, z_3) G_R(z_2, z_2) \\
 \mathcal{T}_{12,7} &= 64ig_2^3 N_c^{-2} G^<(x_1, z_1) G^<(z_2, z_3) G_A(x_2, z_2) G_A(x_4, z_3) G_A(z_2, z_2) G_K(x_3, z_3) G_R(z_1, z_1) G_R(z_1, z_3) \\
 \mathcal{T}_{12,8} &= 64ig_2^3 N_c^{-2} G^<(x_1, z_1) G^<(z_2, z_3) G_A(x_2, z_2) G_A(x_4, z_3) G_A(z_2, z_2) G_K(z_1, z_3) G_R(x_3, z_3) G_R(z_1, z_1) \\
 \mathcal{T}_{12,9} &= 64ig_2^3 N_c^{-2} G^<(x_1, z_1) G^<(z_2, z_3) G_A(x_2, z_2) G_A(x_4, z_3) G_K(x_3, z_3) G_R(z_1, z_1) G_R(z_1, z_3) G_R(z_2, z_2) \\
 \mathcal{T}_{12,10} &= 64ig_2^3 N_c^{-2} G^<(x_1, z_1) G^<(z_2, z_3) G_A(x_2, z_2) G_A(x_4, z_3) G_K(z_1, z_3) G_R(x_3, z_3) G_R(z_1, z_1) G_R(z_2, z_2) \\
 \mathcal{T}_{12,11} &= 64ig_2^3 N_c^{-2} G^<(z_1, z_3) G^<(z_2, z_3) G_A(x_2, z_2) G_A(x_4, z_3) G_A(z_1, z_1) G_A(z_2, z_2) G_K(x_1, z_1) G_R(x_3, z_3) \\
 \mathcal{T}_{12,12} &= 64ig_2^3 N_c^{-2} G^<(z_1, z_3) G^<(z_2, z_3) G_A(x_2, z_2) G_A(x_4, z_3) G_A(z_1, z_1) G_K(x_1, z_1) G_R(x_3, z_3) G_R(z_2, z_2) \\
 \mathcal{T}_{12,13} &= 64ig_2^3 N_c^{-2} G^<(z_1, z_3) G^<(z_2, z_3) G_A(x_2, z_2) G_A(x_4, z_3) G_A(z_2, z_2) G_K(x_1, z_1) G_R(x_3, z_3) G_R(z_1, z_1) \\
 \mathcal{T}_{12,14} &= 64ig_2^3 N_c^{-2} G^<(z_1, z_3) G^<(z_2, z_3) G_A(x_2, z_2) G_A(x_4, z_3) G_A(z_2, z_2) G_K(z_1, z_1) G_R(x_1, z_1) G_R(x_3, z_3) \\
 \mathcal{T}_{12,15} &= 64ig_2^3 N_c^{-2} G^<(z_1, z_3) G^<(z_2, z_3) G_A(x_2, z_2) G_A(x_4, z_3) G_K(x_1, z_1) G_R(x_3, z_3) G_R(z_1, z_1) G_R(z_2, z_2) \\
 \mathcal{T}_{12,16} &= 64ig_2^3 N_c^{-2} G^<(z_1, z_3) G^<(z_2, z_3) G_A(x_2, z_2) G_A(x_4, z_3) G_K(z_1, z_1) G_R(x_1, z_1) G_R(x_3, z_3) G_R(z_2, z_2)
 \end{aligned}$$

Topology 13 (FeynArts Topology 4)

2 → 2

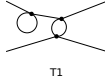


Parsed propagators: 8. Internal vertices: 3. Internal half-edges: 12.
Assignments checked: 32768. Surviving (nonzero vertex weight): 4096.
Raw N_c power: 1. Net power in $C(t)$: -3.

No surviving SK contributions for this topology.

Topology 14 (FeynArts Topology 4)

$2 \rightarrow 2$

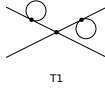


Parsed propagators: 8. Internal vertices: 3. Internal half-edges: 12.
Assignments checked: 32768. Surviving (nonzero vertex weight): 4096.
Raw N_c power: 1. Net power in $C(t)$: -3.

No surviving SK contributions for this topology.

Topology 15 (FeynArts Topology 4)

$2 \rightarrow 2$

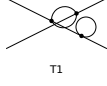


Parsed propagators: 8. Internal vertices: 3. Internal half-edges: 12.
Assignments checked: 32768. Surviving (nonzero vertex weight): 4096.
Raw N_c power: 2. Net power in $C(t)$: -2.

No surviving SK contributions for this topology.

Topology 16 (FeynArts Topology 4)

2 → 2

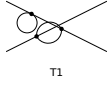


Parsed propagators: 8. Internal vertices: 3. Internal half-edges: 12.
Assignments checked: 32768. Surviving (nonzero vertex weight): 4096.
Raw N_c power: 1. Net power in $C(t)$: -3. Explicit surviving terms:

$$\begin{aligned}\mathcal{T}_{16,1} &= 128ig_2^3 N_c^{-3} G^>(x_3, z_2) G^>(z_3, z_1) G_A(x_2, z_2) G_A(x_4, z_3) G_A(z_1, z_2) G_A(z_3, z_3) G_K(x_1, z_1) G_R(z_1, z_2) \\ \mathcal{T}_{16,2} &= 128ig_2^3 N_c^{-3} G^>(x_3, z_2) G^>(z_3, z_1) G_A(x_2, z_2) G_A(x_4, z_3) G_A(z_1, z_2) G_K(x_1, z_1) G_R(z_1, z_2) G_R(z_3, z_3) \\ \mathcal{T}_{16,3} &= 128ig_2^3 N_c^{-3} G^>(x_3, z_2) G^>(z_3, z_1) G_A(x_2, z_2) G_A(x_4, z_3) G_A(z_3, z_3) G_K(z_1, z_2) G_R(x_1, z_1) G_R(z_1, z_2) \\ \mathcal{T}_{16,4} &= 128ig_2^3 N_c^{-3} G^>(x_3, z_2) G^>(z_3, z_1) G_A(x_2, z_2) G_A(x_4, z_3) G_K(z_1, z_2) G_R(x_1, z_1) G_R(z_1, z_2) G_R(z_3, z_3)\end{aligned}$$

Topology 17 (FeynArts Topology 4)

2 → 2

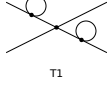


Parsed propagators: 8. Internal vertices: 3. Internal half-edges: 12.
Assignments checked: 32768. Surviving (nonzero vertex weight): 4096.
Raw N_c power: 1. Net power in $C(t)$: -3.

No surviving SK contributions for this topology.

Topology 18 (FeynArts Topology 4)

2 → 2

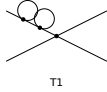


Parsed propagators: 8. Internal vertices: 3. Internal half-edges: 12.
 Assignments checked: 32768. Surviving (nonzero vertex weight): 4096.
 Raw N_c power: 2. Net power in $C(t)$: -2. Explicit surviving terms:

$$\begin{aligned}\mathcal{T}_{18,1} &= 64ig_2^3 N_c^{-2} G^>(x_3, z_2) G^>(z_3, z_2) G_A(x_2, z_2) G_A(x_4, z_3) G_A(z_1, z_1) G_A(z_3, z_3) G_K(x_1, z_1) G_R(z_1, z_2) \\ \mathcal{T}_{18,2} &= 64ig_2^3 N_c^{-2} G^>(x_3, z_2) G^>(z_3, z_2) G_A(x_2, z_2) G_A(x_4, z_3) G_A(z_1, z_1) G_K(x_1, z_1) G_R(z_1, z_2) G_R(z_3, z_3) \\ \mathcal{T}_{18,3} &= 64ig_2^3 N_c^{-2} G^>(x_3, z_2) G^>(z_3, z_2) G_A(x_2, z_2) G_A(x_4, z_3) G_A(z_3, z_3) G_K(x_1, z_1) G_R(z_1, z_1) G_R(z_1, z_2) \\ \mathcal{T}_{18,4} &= 64ig_2^3 N_c^{-2} G^>(x_3, z_2) G^>(z_3, z_2) G_A(x_2, z_2) G_A(x_4, z_3) G_A(z_3, z_3) G_K(z_1, z_1) G_R(x_1, z_1) G_R(z_1, z_2) \\ \mathcal{T}_{18,5} &= 64ig_2^3 N_c^{-2} G^>(x_3, z_2) G^>(z_3, z_2) G_A(x_2, z_2) G_A(x_4, z_3) G_K(x_1, z_1) G_R(z_1, z_1) G_R(z_1, z_2) G_R(z_3, z_3) \\ \mathcal{T}_{18,6} &= 64ig_2^3 N_c^{-2} G^>(x_3, z_2) G^>(z_3, z_2) G_A(x_2, z_2) G_A(x_4, z_3) G_K(z_1, z_1) G_R(x_1, z_1) G_R(z_1, z_2) G_R(z_3, z_3)\end{aligned}$$

Topology 19 (FeynArts Topology 4)

2 → 2

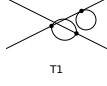


Parsed propagators: 8. Internal vertices: 3. Internal half-edges: 12.
 Assignments checked: 32768. Surviving (nonzero vertex weight): 4096.
 Raw N_c power: 2. Net power in $C(t)$: -2.

No surviving SK contributions for this topology.

Topology 20 (FeynArts Topology 4)

2 → 2

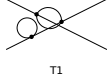


Parsed propagators: 8. Internal vertices: 3. Internal half-edges: 12.
 Assignments checked: 32768. Surviving (nonzero vertex weight): 4096.
 Raw N_c power: 1. Net power in $C(t)$: -3.

No surviving SK contributions for this topology.

Topology 21 (FeynArts Topology 4)

2 → 2

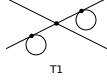


Parsed propagators: 8. Internal vertices: 3. Internal half-edges: 12.
 Assignments checked: 32768. Surviving (nonzero vertex weight): 4096.
 Raw N_c power: 1. Net power in $C(t)$: -3. Explicit surviving terms:

$$\begin{aligned}\mathcal{T}_{21,1} &= 128ig_2^3 N_c^{-3} G^<(x_1, z_1) G^>(z_3, z_2) G_A(x_2, z_2) G_A(x_4, z_1) G_A(z_2, z_2) G_A(z_3, z_1) G_K(x_3, z_3) G_R(z_3, z_1) \\ \mathcal{T}_{21,2} &= 128ig_2^3 N_c^{-3} G^<(x_1, z_1) G^>(z_3, z_2) G_A(x_2, z_2) G_A(x_4, z_1) G_A(z_2, z_2) G_K(z_3, z_1) G_R(x_3, z_3) G_R(z_3, z_1) \\ \mathcal{T}_{21,3} &= 128ig_2^3 N_c^{-3} G^<(x_1, z_1) G^>(z_3, z_2) G_A(x_2, z_2) G_A(x_4, z_1) G_A(z_3, z_1) G_K(x_3, z_3) G_R(z_2, z_2) G_R(z_3, z_1) \\ \mathcal{T}_{21,4} &= 128ig_2^3 N_c^{-3} G^<(x_1, z_1) G^>(z_3, z_2) G_A(x_2, z_2) G_A(x_4, z_1) G_K(z_3, z_1) G_R(x_3, z_3) G_R(z_2, z_2) G_R(z_3, z_1)\end{aligned}$$

Topology 22 (FeynArts Topology 4)

2 → 2

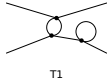


Parsed propagators: 8. Internal vertices: 3. Internal half-edges: 12.
 Assignments checked: 32768. Surviving (nonzero vertex weight): 4096.
 Raw N_c power: 2. Net power in $C(t)$: -2. Explicit surviving terms:

$$\begin{aligned}\mathcal{T}_{22,1} &= 64ig_2^3 N_c^{-2} G^<(x_1, z_1) G^<(z_2, z_1) G_A(x_2, z_2) G_A(x_4, z_1) G_A(z_2, z_2) G_A(z_3, z_3) G_K(x_3, z_3) G_R(z_3, z_1) \\ \mathcal{T}_{22,2} &= 64ig_2^3 N_c^{-2} G^<(x_1, z_1) G^<(z_2, z_1) G_A(x_2, z_2) G_A(x_4, z_1) G_A(z_2, z_2) G_K(x_3, z_3) G_R(z_3, z_1) G_R(z_3, z_3) \\ \mathcal{T}_{22,3} &= 64ig_2^3 N_c^{-2} G^<(x_1, z_1) G^<(z_2, z_1) G_A(x_2, z_2) G_A(x_4, z_1) G_A(z_2, z_2) G_K(z_3, z_3) G_R(x_3, z_3) G_R(z_3, z_1) \\ \mathcal{T}_{22,4} &= 64ig_2^3 N_c^{-2} G^<(x_1, z_1) G^<(z_2, z_1) G_A(x_2, z_2) G_A(x_4, z_1) G_A(z_3, z_3) G_K(x_3, z_3) G_R(z_2, z_2) G_R(z_3, z_1) \\ \mathcal{T}_{22,5} &= 64ig_2^3 N_c^{-2} G^<(x_1, z_1) G^<(z_2, z_1) G_A(x_2, z_2) G_A(x_4, z_1) G_K(x_3, z_3) G_R(z_2, z_2) G_R(z_3, z_1) G_R(z_3, z_3) \\ \mathcal{T}_{22,6} &= 64ig_2^3 N_c^{-2} G^<(x_1, z_1) G^<(z_2, z_1) G_A(x_2, z_2) G_A(x_4, z_1) G_K(z_3, z_3) G_R(x_3, z_3) G_R(z_2, z_2) G_R(z_3, z_1)\end{aligned}$$

Topology 23 (FeynArts Topology 4)

2 → 2

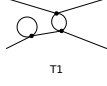


Parsed propagators: 8. Internal vertices: 3. Internal half-edges: 12.
 Assignments checked: 32768. Surviving (nonzero vertex weight): 4096.
 Raw N_c power: 1. Net power in $C(t)$: -3. Explicit surviving terms:

$$\begin{aligned}\mathcal{T}_{23,1} &= 128ig_2^3 N_c^{-3} G^>(x_3, z_1) G^>(z_3, z_2) G_A(x_2, z_2) G_A(x_4, z_3) G_A(z_2, z_1) G_A(z_3, z_3) G_K(x_1, z_1) G_R(z_2, z_1) \\ \mathcal{T}_{23,2} &= 128ig_2^3 N_c^{-3} G^>(x_3, z_1) G^>(z_3, z_2) G_A(x_2, z_2) G_A(x_4, z_3) G_A(z_2, z_1) G_A(z_3, z_3) G_K(z_2, z_1) G_R(x_1, z_1) \\ \mathcal{T}_{23,3} &= 128ig_2^3 N_c^{-3} G^>(x_3, z_1) G^>(z_3, z_2) G_A(x_2, z_2) G_A(x_4, z_3) G_A(z_2, z_1) G_K(x_1, z_1) G_R(z_2, z_1) G_R(z_3, z_3) \\ \mathcal{T}_{23,4} &= 128ig_2^3 N_c^{-3} G^>(x_3, z_1) G^>(z_3, z_2) G_A(x_2, z_2) G_A(x_4, z_3) G_A(z_2, z_1) G_K(z_2, z_1) G_R(x_1, z_1) G_R(z_3, z_3)\end{aligned}$$

Topology 24 (FeynArts Topology 4)

2 → 2



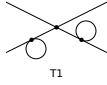
11

Parsed propagators: 8. Internal vertices: 3. Internal half-edges: 12.
 Assignments checked: 32768. Surviving (nonzero vertex weight): 4096.
 Raw N_c power: 1. Net power in $C(t)$: -3. Explicit surviving terms:

$$\begin{aligned}\mathcal{T}_{24,1} &= 128ig_2^3 N_c^{-3} G^<(x_1, z_1) G^>(z_3, z_2) G_A(x_2, z_2) G_A(x_4, z_3) G_A(z_2, z_2) G_A(z_3, z_1) G_K(x_3, z_1) G_R(z_3, z_1) \\ \mathcal{T}_{24,2} &= 128ig_2^3 N_c^{-3} G^<(x_1, z_1) G^>(z_3, z_2) G_A(x_2, z_2) G_A(x_4, z_3) G_A(z_2, z_2) G_A(z_3, z_1) G_K(z_3, z_1) G_R(x_3, z_1) \\ \mathcal{T}_{24,3} &= 128ig_2^3 N_c^{-3} G^<(x_1, z_1) G^>(z_3, z_2) G_A(x_2, z_2) G_A(x_4, z_3) G_A(z_3, z_1) G_K(x_3, z_1) G_R(z_2, z_2) G_R(z_3, z_1) \\ \mathcal{T}_{24,4} &= 128ig_2^3 N_c^{-3} G^<(x_1, z_1) G^>(z_3, z_2) G_A(x_2, z_2) G_A(x_4, z_3) G_A(z_3, z_1) G_K(z_3, z_1) G_R(x_3, z_1) G_R(z_2, z_2)\end{aligned}$$

Topology 25 (FeynArts Topology 4)

2 → 2



11

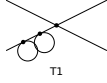
Parsed propagators: 8. Internal vertices: 3. Internal half-edges: 12.
 Assignments checked: 32768. Surviving (nonzero vertex weight): 4096.
 Raw N_c power: 2. Net power in $C(t)$: -2. Explicit surviving terms:

$$\begin{aligned}\mathcal{T}_{25,1} &= 64ig_2^3 N_c^{-2} G^<(x_1, z_1) G^<(z_2, z_1) G_A(x_2, z_2) G_A(x_4, z_3) G_A(z_2, z_2) G_A(z_3, z_1) G_K(z_3, z_3) G_R(x_3, z_1) \\ \mathcal{T}_{25,2} &= 64ig_2^3 N_c^{-2} G^<(x_1, z_1) G^<(z_2, z_1) G_A(x_2, z_2) G_A(x_4, z_3) G_A(z_2, z_2) G_A(z_3, z_3) G_K(x_3, z_1) G_R(z_3, z_1) \\ \mathcal{T}_{25,3} &= 64ig_2^3 N_c^{-2} G^<(x_1, z_1) G^<(z_2, z_1) G_A(x_2, z_2) G_A(x_4, z_3) G_A(z_2, z_2) G_A(z_3, z_3) G_K(z_3, z_1) G_R(x_3, z_1) \\ \mathcal{T}_{25,4} &= 64ig_2^3 N_c^{-2} G^<(x_1, z_1) G^<(z_2, z_1) G_A(x_2, z_2) G_A(x_4, z_3) G_A(z_2, z_2) G_K(x_3, z_1) G_R(z_3, z_1) G_R(z_3, z_3) \\ \mathcal{T}_{25,5} &= 64ig_2^3 N_c^{-2} G^<(x_1, z_1) G^<(z_2, z_1) G_A(x_2, z_2) G_A(x_4, z_3) G_A(z_2, z_2) G_K(z_3, z_1) G_R(x_3, z_1) G_R(z_3, z_3) \\ \mathcal{T}_{25,6} &= 64ig_2^3 N_c^{-2} G^<(x_1, z_1) G^<(z_2, z_1) G_A(x_2, z_2) G_A(x_4, z_3) G_A(z_3, z_1) G_K(z_3, z_3) G_R(x_3, z_1) G_R(z_2, z_2) \\ \mathcal{T}_{25,7} &= 64ig_2^3 N_c^{-2} G^<(x_1, z_1) G^<(z_2, z_1) G_A(x_2, z_2) G_A(x_4, z_3) G_A(z_3, z_3) G_K(x_3, z_1) G_R(z_2, z_2) G_R(z_3, z_1) \\ \mathcal{T}_{25,8} &= 64ig_2^3 N_c^{-2} G^<(x_1, z_1) G^<(z_2, z_1) G_A(x_2, z_2) G_A(x_4, z_3) G_A(z_3, z_3) G_K(z_3, z_1) G_R(x_3, z_1) G_R(z_2, z_2)\end{aligned}$$

$$\begin{aligned}
\mathcal{T}_{25,9} &= 64ig_2^3 N_c^{-2} G^<(x_1, z_1) G^<(z_2, z_1) G_A(x_2, z_2) G_A(x_4, z_3) G_K(x_3, z_1) G_R(z_2, z_2) G_R(z_3, z_1) G_R(z_3, z_3) \\
\mathcal{T}_{25,10} &= 64ig_2^3 N_c^{-2} G^<(x_1, z_1) G^<(z_2, z_1) G_A(x_2, z_2) G_A(x_4, z_3) G_K(z_3, z_1) G_R(x_3, z_1) G_R(z_2, z_2) G_R(z_3, z_3) \\
\mathcal{T}_{25,11} &= 64ig_2^3 N_c^{-2} G^>(x_3, z_1) G^>(z_3, z_1) G_A(x_2, z_2) G_A(x_4, z_3) G_A(z_2, z_1) G_A(z_3, z_3) G_K(z_2, z_2) G_R(x_1, z_1) \\
\mathcal{T}_{25,12} &= 64ig_2^3 N_c^{-2} G^>(x_3, z_1) G^>(z_3, z_1) G_A(x_2, z_2) G_A(x_4, z_3) G_A(z_2, z_1) G_K(z_2, z_2) G_R(x_1, z_1) G_R(z_3, z_3) \\
\mathcal{T}_{25,13} &= 64ig_2^3 N_c^{-2} G^>(x_3, z_1) G^>(z_3, z_1) G_A(x_2, z_2) G_A(x_4, z_3) G_A(z_2, z_2) G_A(z_3, z_3) G_K(x_1, z_1) G_R(z_2, z_1) \\
\mathcal{T}_{25,14} &= 64ig_2^3 N_c^{-2} G^>(x_3, z_1) G^>(z_3, z_1) G_A(x_2, z_2) G_A(x_4, z_3) G_A(z_2, z_2) G_A(z_3, z_3) G_K(z_2, z_1) G_R(x_1, z_1) \\
\mathcal{T}_{25,15} &= 64ig_2^3 N_c^{-2} G^>(x_3, z_1) G^>(z_3, z_1) G_A(x_2, z_2) G_A(x_4, z_3) G_A(z_2, z_2) G_K(x_1, z_1) G_R(z_2, z_1) G_R(z_3, z_3) \\
\mathcal{T}_{25,16} &= 64ig_2^3 N_c^{-2} G^>(x_3, z_1) G^>(z_3, z_1) G_A(x_2, z_2) G_A(x_4, z_3) G_A(z_2, z_2) G_K(z_2, z_1) G_R(x_1, z_1) G_R(z_3, z_3) \\
\mathcal{T}_{25,17} &= 64ig_2^3 N_c^{-2} G^>(x_3, z_1) G^>(z_3, z_1) G_A(x_2, z_2) G_A(x_4, z_3) G_A(z_3, z_3) G_K(x_1, z_1) G_R(z_2, z_1) G_R(z_2, z_2) \\
\mathcal{T}_{25,18} &= 64ig_2^3 N_c^{-2} G^>(x_3, z_1) G^>(z_3, z_1) G_A(x_2, z_2) G_A(x_4, z_3) G_A(z_3, z_3) G_K(z_2, z_1) G_R(x_1, z_1) G_R(z_2, z_2) \\
\mathcal{T}_{25,19} &= 64ig_2^3 N_c^{-2} G^>(x_3, z_1) G^>(z_3, z_1) G_A(x_2, z_2) G_A(x_4, z_3) G_K(x_1, z_1) G_R(z_2, z_1) G_R(z_2, z_2) G_R(z_3, z_3) \\
\mathcal{T}_{25,20} &= 64ig_2^3 N_c^{-2} G^>(x_3, z_1) G^>(z_3, z_1) G_A(x_2, z_2) G_A(x_4, z_3) G_K(z_2, z_1) G_R(x_1, z_1) G_R(z_2, z_2) G_R(z_3, z_3)
\end{aligned}$$

Topology 26 (FeynArts Topology 4)

2 → 2



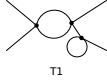
Parsed propagators: 8. Internal vertices: 3. Internal half-edges: 12.
Assignments checked: 32768. Surviving (nonzero vertex weight): 4096.
Raw N_c power: 2. Net power in $C(t)$: -2. Explicit surviving terms:

$$\begin{aligned}
\mathcal{T}_{26,1} &= 64ig_2^3 N_c^{-2} G^<(x_1, z_1) G^<(z_3, z_1) G_A(x_2, z_2) G_A(x_4, z_1) G_A(z_2, z_2) G_A(z_3, z_2) G_K(z_3, z_3) G_R(x_3, z_1) \\
\mathcal{T}_{26,2} &= 64ig_2^3 N_c^{-2} G^<(x_1, z_1) G^<(z_3, z_1) G_A(x_2, z_2) G_A(x_4, z_1) G_A(z_2, z_2) G_A(z_3, z_3) G_K(z_3, z_2) G_R(x_3, z_1) \\
\mathcal{T}_{26,3} &= 64ig_2^3 N_c^{-2} G^<(x_1, z_1) G^<(z_3, z_1) G_A(x_2, z_2) G_A(x_4, z_1) G_A(z_2, z_2) G_K(z_3, z_2) G_R(x_3, z_1) G_R(z_3, z_3) \\
\mathcal{T}_{26,4} &= 64ig_2^3 N_c^{-2} G^<(x_1, z_1) G^<(z_3, z_1) G_A(x_2, z_2) G_A(x_4, z_1) G_A(z_3, z_2) G_K(z_3, z_3) G_R(x_3, z_1) G_R(z_2, z_2) \\
\mathcal{T}_{26,5} &= 64ig_2^3 N_c^{-2} G^<(x_1, z_1) G^<(z_3, z_1) G_A(x_2, z_2) G_A(x_4, z_1) G_A(z_3, z_3) G_K(z_2, z_2) G_R(x_3, z_1) G_R(z_3, z_2) \\
\mathcal{T}_{26,6} &= 64ig_2^3 N_c^{-2} G^<(x_1, z_1) G^<(z_3, z_1) G_A(x_2, z_2) G_A(x_4, z_1) G_A(z_3, z_3) G_K(z_3, z_2) G_R(x_3, z_1) G_R(z_2, z_2) \\
\mathcal{T}_{26,7} &= 64ig_2^3 N_c^{-2} G^<(x_1, z_1) G^<(z_3, z_1) G_A(x_2, z_2) G_A(x_4, z_1) G_K(z_2, z_2) G_R(x_3, z_1) G_R(z_3, z_2) G_R(z_3, z_3) \\
\mathcal{T}_{26,8} &= 64ig_2^3 N_c^{-2} G^<(x_1, z_1) G^<(z_3, z_1) G_A(x_2, z_2) G_A(x_4, z_1) G_K(z_3, z_2) G_R(x_3, z_1) G_R(z_2, z_2) G_R(z_3, z_3) \\
\mathcal{T}_{26,9} &= 64ig_2^3 N_c^{-2} G^<(x_1, z_1) G^>(z_3, z_2) G_A(x_2, z_2) G_A(x_4, z_1) G_A(z_2, z_2) G_A(z_3, z_1) G_K(z_3, z_3) G_R(x_3, z_1) \\
\mathcal{T}_{26,10} &= 64ig_2^3 N_c^{-2} G^<(x_1, z_1) G^>(z_3, z_2) G_A(x_2, z_2) G_A(x_4, z_1) G_A(z_2, z_2) G_A(z_3, z_3) G_K(x_3, z_1) G_R(z_3, z_1) \\
\mathcal{T}_{26,11} &= 64ig_2^3 N_c^{-2} G^<(x_1, z_1) G^>(z_3, z_2) G_A(x_2, z_2) G_A(x_4, z_1) G_A(z_2, z_2) G_A(z_3, z_3) G_K(z_3, z_1) G_R(x_3, z_1) \\
\mathcal{T}_{26,12} &= 64ig_2^3 N_c^{-2} G^<(x_1, z_1) G^>(z_3, z_2) G_A(x_2, z_2) G_A(x_4, z_1) G_A(z_2, z_2) G_K(x_3, z_1) G_R(z_3, z_1) G_R(z_3, z_3) \\
\mathcal{T}_{26,13} &= 64ig_2^3 N_c^{-2} G^<(x_1, z_1) G^>(z_3, z_2) G_A(x_2, z_2) G_A(x_4, z_1) G_A(z_2, z_2) G_K(z_3, z_1) G_R(x_3, z_1) G_R(z_3, z_3)
\end{aligned}$$

$$\begin{aligned}
\mathcal{T}_{26,14} &= 64ig_2^3 N_c^{-2} G^<(x_1, z_1) G^>(z_3, z_2) G_A(x_2, z_2) G_A(x_4, z_1) G_A(z_3, z_1) G_K(z_3, z_3) G_R(x_3, z_1) G_R(z_2, z_2) \\
\mathcal{T}_{26,15} &= 64ig_2^3 N_c^{-2} G^<(x_1, z_1) G^>(z_3, z_2) G_A(x_2, z_2) G_A(x_4, z_1) G_A(z_3, z_3) G_K(x_3, z_1) G_R(z_2, z_2) G_R(z_3, z_1) \\
\mathcal{T}_{26,16} &= 64ig_2^3 N_c^{-2} G^<(x_1, z_1) G^>(z_3, z_2) G_A(x_2, z_2) G_A(x_4, z_1) G_A(z_3, z_3) G_K(z_3, z_1) G_R(x_3, z_1) G_R(z_2, z_2) \\
\mathcal{T}_{26,17} &= 64ig_2^3 N_c^{-2} G^<(x_1, z_1) G^>(z_3, z_2) G_A(x_2, z_2) G_A(x_4, z_1) G_K(x_3, z_1) G_R(z_2, z_2) G_R(z_3, z_1) G_R(z_3, z_3) \\
\mathcal{T}_{26,18} &= 64ig_2^3 N_c^{-2} G^<(x_1, z_1) G^>(z_3, z_2) G_A(x_2, z_2) G_A(x_4, z_1) G_K(z_3, z_1) G_R(x_3, z_1) G_R(z_2, z_2) G_R(z_3, z_3)
\end{aligned}$$

Topology 27 (FeynArts Topology 4)

$$2 \rightarrow 2$$

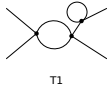


Parsed propagators: 8. Internal vertices: 3. Internal half-edges: 12.
Assignments checked: 32768. Surviving (nonzero vertex weight): 4096.
Raw N_c power: 1. Net power in $C(t)$: -3. Explicit surviving terms:

$$\begin{aligned}
\mathcal{T}_{27,1} &= 128ig_2^3 N_c^{-3} G^>(x_3, z_2) G^>(z_3, z_2) G_A(x_2, z_1) G_A(x_4, z_3) G_A(z_2, z_1) G_A(z_3, z_3) G_K(x_1, z_1) G_R(z_2, z_1) \\
\mathcal{T}_{27,2} &= 128ig_2^3 N_c^{-3} G^>(x_3, z_2) G^>(z_3, z_2) G_A(x_2, z_1) G_A(x_4, z_3) G_A(z_2, z_1) G_A(z_3, z_3) G_K(z_2, z_1) G_R(x_1, z_1) \\
\mathcal{T}_{27,3} &= 128ig_2^3 N_c^{-3} G^>(x_3, z_2) G^>(z_3, z_2) G_A(x_2, z_1) G_A(x_4, z_3) G_A(z_2, z_1) G_K(x_1, z_1) G_R(z_2, z_1) G_R(z_3, z_3) \\
\mathcal{T}_{27,4} &= 128ig_2^3 N_c^{-3} G^>(x_3, z_2) G^>(z_3, z_2) G_A(x_2, z_1) G_A(x_4, z_3) G_A(z_2, z_1) G_K(z_2, z_1) G_R(x_1, z_1) G_R(z_3, z_3) \\
\mathcal{T}_{27,5} &= 64ig_2^3 N_c^{-3} G^>(z_2, z_1) G^>(z_2, z_1) G_A(x_2, z_1) G_A(x_4, z_3) G_A(z_3, z_2) G_K(z_3, z_3) G_R(x_1, z_1) G_R(x_3, z_2) \\
\mathcal{T}_{27,6} &= 64ig_2^3 N_c^{-3} G^>(z_2, z_1) G^>(z_2, z_1) G_A(x_2, z_1) G_A(x_4, z_3) G_A(z_3, z_3) G_K(x_3, z_2) G_R(x_1, z_1) G_R(z_3, z_2) \\
\mathcal{T}_{27,7} &= 64ig_2^3 N_c^{-3} G^>(z_2, z_1) G^>(z_2, z_1) G_A(x_2, z_1) G_A(x_4, z_3) G_A(z_3, z_3) G_K(z_3, z_2) G_R(x_1, z_1) G_R(x_3, z_2) \\
\mathcal{T}_{27,8} &= 64ig_2^3 N_c^{-3} G^>(z_2, z_1) G^>(z_2, z_1) G_A(x_2, z_1) G_A(x_4, z_3) G_K(x_3, z_2) G_R(x_1, z_1) G_R(z_3, z_2) G_R(z_3, z_3) \\
\mathcal{T}_{27,9} &= 64ig_2^3 N_c^{-3} G^>(z_2, z_1) G^>(z_2, z_1) G_A(x_2, z_1) G_A(x_4, z_3) G_K(z_3, z_2) G_R(x_1, z_1) G_R(x_3, z_2) G_R(z_3, z_3)
\end{aligned}$$

Topology 28 (FeynArts Topology 4)

$$2 \rightarrow 2$$

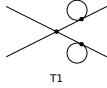


Parsed propagators: 8. Internal vertices: 3. Internal half-edges: 12.
Assignments checked: 32768. Surviving (nonzero vertex weight): 4096.
Raw N_c power: 1. Net power in $C(t)$: -3. Explicit surviving terms:

$$\begin{aligned}\mathcal{T}_{28,1} &= 64ig_2^3 N_c^{-3} G^>(z_3, z_1) G^>(z_3, z_1) G_A(x_2, z_1) G_A(x_4, z_3) G_A(z_2, z_2) G_A(z_3, z_2) G_K(x_3, z_2) G_R(x_1, z_1) \\ \mathcal{T}_{28,2} &= 64ig_2^3 N_c^{-3} G^>(z_3, z_1) G^>(z_3, z_1) G_A(x_2, z_1) G_A(x_4, z_3) G_A(z_3, z_2) G_K(x_3, z_2) G_R(x_1, z_1) G_R(z_2, z_2) \\ \mathcal{T}_{28,3} &= 64ig_2^3 N_c^{-3} G^>(z_3, z_1) G^>(z_3, z_1) G_A(x_2, z_1) G_A(x_4, z_3) G_A(z_3, z_2) G_K(z_2, z_2) G_R(x_1, z_1) G_R(x_3, z_2)\end{aligned}$$

Topology 29 (FeynArts Topology 4)

2 → 2

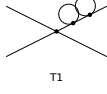


Parsed propagators: 8. Internal vertices: 3. Internal half-edges: 12.
Assignments checked: 32768. Surviving (nonzero vertex weight): 4096.
Raw N_c power: 2. Net power in $C(t)$: -2. Explicit surviving terms:

$$\begin{aligned}\mathcal{T}_{29,1} &= 64ig_2^3 N_c^{-2} G^>(x_3, z_2) G^>(z_3, z_1) G_A(x_2, z_1) G_A(x_4, z_3) G_A(z_2, z_1) G_A(z_3, z_3) G_K(z_2, z_2) G_R(x_1, z_1) \\ \mathcal{T}_{29,2} &= 64ig_2^3 N_c^{-2} G^>(x_3, z_2) G^>(z_3, z_1) G_A(x_2, z_1) G_A(x_4, z_3) G_A(z_2, z_1) G_K(z_2, z_2) G_R(x_1, z_1) G_R(z_3, z_3) \\ \mathcal{T}_{29,3} &= 64ig_2^3 N_c^{-2} G^>(x_3, z_2) G^>(z_3, z_1) G_A(x_2, z_1) G_A(x_4, z_3) G_A(z_2, z_2) G_A(z_3, z_3) G_K(x_1, z_1) G_R(z_2, z_1) \\ \mathcal{T}_{29,4} &= 64ig_2^3 N_c^{-2} G^>(x_3, z_2) G^>(z_3, z_1) G_A(x_2, z_1) G_A(x_4, z_3) G_A(z_2, z_2) G_A(z_3, z_3) G_K(z_2, z_1) G_R(x_1, z_1) \\ \mathcal{T}_{29,5} &= 64ig_2^3 N_c^{-2} G^>(x_3, z_2) G^>(z_3, z_1) G_A(x_2, z_1) G_A(x_4, z_3) G_A(z_2, z_2) G_K(x_1, z_1) G_R(z_2, z_1) G_R(z_3, z_3) \\ \mathcal{T}_{29,6} &= 64ig_2^3 N_c^{-2} G^>(x_3, z_2) G^>(z_3, z_1) G_A(x_2, z_1) G_A(x_4, z_3) G_A(z_2, z_2) G_K(z_2, z_1) G_R(x_1, z_1) G_R(z_3, z_3) \\ \mathcal{T}_{29,7} &= 64ig_2^3 N_c^{-2} G^>(x_3, z_2) G^>(z_3, z_1) G_A(x_2, z_1) G_A(x_4, z_3) G_A(z_3, z_3) G_K(x_1, z_1) G_R(z_2, z_1) G_R(z_2, z_2) \\ \mathcal{T}_{29,8} &= 64ig_2^3 N_c^{-2} G^>(x_3, z_2) G^>(z_3, z_1) G_A(x_2, z_1) G_A(x_4, z_3) G_A(z_3, z_3) G_K(z_2, z_1) G_R(x_1, z_1) G_R(z_2, z_2) \\ \mathcal{T}_{29,9} &= 64ig_2^3 N_c^{-2} G^>(x_3, z_2) G^>(z_3, z_1) G_A(x_2, z_1) G_A(x_4, z_3) G_K(x_1, z_1) G_R(z_2, z_1) G_R(z_2, z_2) G_R(z_3, z_3) \\ \mathcal{T}_{29,10} &= 64ig_2^3 N_c^{-2} G^>(x_3, z_2) G^>(z_3, z_1) G_A(x_2, z_1) G_A(x_4, z_3) G_K(z_2, z_1) G_R(x_1, z_1) G_R(z_2, z_2) G_R(z_3, z_3) \\ \mathcal{T}_{29,11} &= 64ig_2^3 N_c^{-2} G^>(z_2, z_1) G^>(z_3, z_1) G_A(x_2, z_1) G_A(x_4, z_3) G_A(z_2, z_2) G_A(z_3, z_3) G_K(x_3, z_2) G_R(x_1, z_1) \\ \mathcal{T}_{29,12} &= 64ig_2^3 N_c^{-2} G^>(z_2, z_1) G^>(z_3, z_1) G_A(x_2, z_1) G_A(x_4, z_3) G_A(z_2, z_2) G_K(x_3, z_2) G_R(x_1, z_1) G_R(z_3, z_3) \\ \mathcal{T}_{29,13} &= 64ig_2^3 N_c^{-2} G^>(z_2, z_1) G^>(z_3, z_1) G_A(x_2, z_1) G_A(x_4, z_3) G_A(z_3, z_3) G_K(x_3, z_2) G_R(x_1, z_1) G_R(z_2, z_2) \\ \mathcal{T}_{29,14} &= 64ig_2^3 N_c^{-2} G^>(z_2, z_1) G^>(z_3, z_1) G_A(x_2, z_1) G_A(x_4, z_3) G_A(z_3, z_3) G_K(z_2, z_2) G_R(x_1, z_1) G_R(x_3, z_2) \\ \mathcal{T}_{29,15} &= 64ig_2^3 N_c^{-2} G^>(z_2, z_1) G^>(z_3, z_1) G_A(x_2, z_1) G_A(x_4, z_3) G_K(x_3, z_2) G_R(x_1, z_1) G_R(z_2, z_2) G_R(z_3, z_3) \\ \mathcal{T}_{29,16} &= 64ig_2^3 N_c^{-2} G^>(z_2, z_1) G^>(z_3, z_1) G_A(x_2, z_1) G_A(x_4, z_3) G_K(z_2, z_2) G_R(x_1, z_1) G_R(x_3, z_2) G_R(z_3, z_3)\end{aligned}$$

Topology 30 (FeynArts Topology 4)

2 → 2

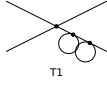


Parsed propagators: 8. Internal vertices: 3. Internal half-edges: 12.
 Assignments checked: 32768. Surviving (nonzero vertex weight): 4096.
 Raw N_c power: 2. Net power in $C(t)$: -2.

No surviving SK contributions for this topology.

Topology 31 (FeynArts Topology 4)

2 → 2



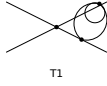
Parsed propagators: 8. Internal vertices: 3. Internal half-edges: 12.
 Assignments checked: 32768. Surviving (nonzero vertex weight): 4096.
 Raw N_c power: 2. Net power in $C(t)$: -2. Explicit surviving terms:

$$\begin{aligned}
 \mathcal{T}_{31,1} &= 64ig_2^3 N_c^{-2} G^<(z_3, z_2) G^>(x_3, z_1) G_A(x_2, z_1) G_A(x_4, z_2) G_A(z_2, z_2) G_A(z_3, z_1) G_K(z_3, z_3) G_R(x_1, z_1) \\
 \mathcal{T}_{31,2} &= 64ig_2^3 N_c^{-2} G^<(z_3, z_2) G^>(x_3, z_1) G_A(x_2, z_1) G_A(x_4, z_2) G_A(z_2, z_2) G_A(z_3, z_3) G_K(x_1, z_1) G_R(z_3, z_1) \\
 \mathcal{T}_{31,3} &= 64ig_2^3 N_c^{-2} G^<(z_3, z_2) G^>(x_3, z_1) G_A(x_2, z_1) G_A(x_4, z_2) G_A(z_2, z_2) G_A(z_3, z_3) G_K(z_3, z_1) G_R(x_1, z_1) \\
 \mathcal{T}_{31,4} &= 64ig_2^3 N_c^{-2} G^<(z_3, z_2) G^>(x_3, z_1) G_A(x_2, z_1) G_A(x_4, z_2) G_A(z_2, z_2) G_K(x_1, z_1) G_R(z_3, z_1) G_R(z_3, z_3) \\
 \mathcal{T}_{31,5} &= 64ig_2^3 N_c^{-2} G^<(z_3, z_2) G^>(x_3, z_1) G_A(x_2, z_1) G_A(x_4, z_2) G_A(z_2, z_2) G_K(z_3, z_1) G_R(x_1, z_1) G_R(z_3, z_3) \\
 \mathcal{T}_{31,6} &= 64ig_2^3 N_c^{-2} G^<(z_3, z_2) G^>(x_3, z_1) G_A(x_2, z_1) G_A(x_4, z_2) G_A(z_3, z_1) G_K(z_3, z_3) G_R(x_1, z_1) G_R(z_2, z_2) \\
 \mathcal{T}_{31,7} &= 64ig_2^3 N_c^{-2} G^<(z_3, z_2) G^>(x_3, z_1) G_A(x_2, z_1) G_A(x_4, z_2) G_A(z_3, z_3) G_K(x_1, z_1) G_R(z_2, z_2) G_R(z_3, z_1) \\
 \mathcal{T}_{31,8} &= 64ig_2^3 N_c^{-2} G^<(z_3, z_2) G^>(x_3, z_1) G_A(x_2, z_1) G_A(x_4, z_2) G_A(z_3, z_3) G_K(z_3, z_1) G_R(x_1, z_1) G_R(z_2, z_2) \\
 \mathcal{T}_{31,9} &= 64ig_2^3 N_c^{-2} G^<(z_3, z_2) G^>(x_3, z_1) G_A(x_2, z_1) G_A(x_4, z_2) G_K(x_1, z_1) G_R(z_2, z_2) G_R(z_3, z_1) G_R(z_3, z_3) \\
 \mathcal{T}_{31,10} &= 64ig_2^3 N_c^{-2} G^<(z_3, z_2) G^>(x_3, z_1) G_A(x_2, z_1) G_A(x_4, z_2) G_K(z_3, z_1) G_R(x_1, z_1) G_R(z_2, z_2) G_R(z_3, z_3) \\
 \mathcal{T}_{31,11} &= 64ig_2^3 N_c^{-2} G^>(x_3, z_1) G^>(z_3, z_1) G_A(x_2, z_1) G_A(x_4, z_2) G_A(z_2, z_2) G_A(z_3, z_2) G_K(z_3, z_3) G_R(x_1, z_1)
 \end{aligned}$$

$$\begin{aligned}
\mathcal{T}_{31,12} &= 64ig_2^3 N_c^{-2} G^>(x_3, z_1) G^>(z_3, z_1) G_A(x_2, z_1) G_A(x_4, z_2) G_A(z_2, z_2) G_A(z_3, z_3) G_K(z_3, z_2) G_R(x_1, z_1) \\
\mathcal{T}_{31,13} &= 64ig_2^3 N_c^{-2} G^>(x_3, z_1) G^>(z_3, z_1) G_A(x_2, z_1) G_A(x_4, z_2) G_A(z_2, z_2) G_K(z_3, z_2) G_R(x_1, z_1) G_R(z_3, z_3) \\
\mathcal{T}_{31,14} &= 64ig_2^3 N_c^{-2} G^>(x_3, z_1) G^>(z_3, z_1) G_A(x_2, z_1) G_A(x_4, z_2) G_A(z_3, z_2) G_K(z_3, z_3) G_R(x_1, z_1) G_R(z_2, z_2) \\
\mathcal{T}_{31,15} &= 64ig_2^3 N_c^{-2} G^>(x_3, z_1) G^>(z_3, z_1) G_A(x_2, z_1) G_A(x_4, z_2) G_A(z_3, z_3) G_K(z_2, z_2) G_R(x_1, z_1) G_R(z_3, z_2) \\
\mathcal{T}_{31,16} &= 64ig_2^3 N_c^{-2} G^>(x_3, z_1) G^>(z_3, z_1) G_A(x_2, z_1) G_A(x_4, z_2) G_A(z_3, z_3) G_K(z_3, z_2) G_R(x_1, z_1) G_R(z_2, z_2) \\
\mathcal{T}_{31,17} &= 64ig_2^3 N_c^{-2} G^>(x_3, z_1) G^>(z_3, z_1) G_A(x_2, z_1) G_A(x_4, z_2) G_K(z_2, z_2) G_R(x_1, z_1) G_R(z_3, z_2) G_R(z_3, z_3) \\
\mathcal{T}_{31,18} &= 64ig_2^3 N_c^{-2} G^>(x_3, z_1) G^>(z_3, z_1) G_A(x_2, z_1) G_A(x_4, z_2) G_K(z_3, z_2) G_R(x_1, z_1) G_R(z_2, z_2) G_R(z_3, z_3)
\end{aligned}$$

Topology 32 (FeynArts Topology 4)

$$2 \rightarrow 2$$

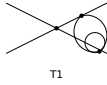


Parsed propagators: 8. Internal vertices: 3. Internal half-edges: 12.
Assignments checked: 32768. Surviving (nonzero vertex weight): 4096.
Raw N_c power: 2. Net power in $C(t)$: -2. Explicit surviving terms:

$$\begin{aligned}
\mathcal{T}_{32,1} &= 128ig_2^3 N_c^{-2} G^<(z_1, z_2) G^>(x_3, z_1) G_A(x_2, z_1) G_A(x_4, z_2) G_A(z_2, z_3) G_A(z_3, z_3) G_K(z_2, z_3) G_R(x_1, z_1) \\
\mathcal{T}_{32,2} &= 128ig_2^3 N_c^{-2} G^<(z_1, z_2) G^>(x_3, z_1) G_A(x_2, z_1) G_A(x_4, z_2) G_A(z_2, z_3) G_K(z_2, z_3) G_R(x_1, z_1) G_R(z_3, z_3) \\
\mathcal{T}_{32,3} &= 128ig_2^3 N_c^{-2} G^<(z_1, z_2) G^>(x_3, z_1) G_A(x_2, z_1) G_A(x_4, z_2) G_A(z_2, z_3) G_K(z_3, z_3) G_R(x_1, z_1) G_R(z_2, z_3)
\end{aligned}$$

Topology 33 (FeynArts Topology 4)

$$2 \rightarrow 2$$

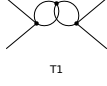


Parsed propagators: 8. Internal vertices: 3. Internal half-edges: 12.
Assignments checked: 32768. Surviving (nonzero vertex weight): 4096.
Raw N_c power: 2. Net power in $C(t)$: -2.

No surviving SK contributions for this topology.

Topology 34 (FeynArts Topology 4)

2 → 2



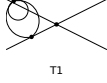
11

Parsed propagators: 8. Internal vertices: 3. Internal half-edges: 12.
 Assignments checked: 32768. Surviving (nonzero vertex weight): 4096.
 Raw N_c power: 1. Net power in $C(t)$: -3. Explicit surviving terms:

$$\begin{aligned}\mathcal{T}_{34,1} &= 128ig_2^3 N_c^{-3} G^<(z_1, z_3) G^<(z_1, z_3) G_A(x_2, z_1) G_A(x_4, z_2) G_A(z_2, z_3) G_K(x_3, z_2) G_R(x_1, z_1) G_R(z_2, z_3) \\ \mathcal{T}_{34,2} &= 128ig_2^3 N_c^{-3} G^<(z_1, z_3) G^<(z_1, z_3) G_A(x_2, z_1) G_A(x_4, z_2) G_K(z_2, z_3) G_R(x_1, z_1) G_R(x_3, z_2) G_R(z_2, z_3) \\ \mathcal{T}_{34,3} &= 128ig_2^3 N_c^{-3} G^>(z_2, z_3) G^>(z_2, z_3) G_A(x_2, z_1) G_A(x_4, z_2) G_A(z_1, z_3) G_K(x_1, z_1) G_R(x_3, z_2) G_R(z_1, z_3) \\ \mathcal{T}_{34,4} &= 128ig_2^3 N_c^{-3} G^>(z_2, z_3) G^>(z_2, z_3) G_A(x_2, z_1) G_A(x_4, z_2) G_K(z_1, z_3) G_R(x_1, z_1) G_R(x_3, z_2) G_R(z_1, z_3)\end{aligned}$$

Topology 35 (FeynArts Topology 4)

2 → 2



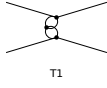
11

Parsed propagators: 8. Internal vertices: 3. Internal half-edges: 12.
 Assignments checked: 32768. Surviving (nonzero vertex weight): 4096.
 Raw N_c power: 2. Net power in $C(t)$: -2. Explicit surviving terms:

$$\begin{aligned}\mathcal{T}_{35,1} &= 128ig_2^3 N_c^{-2} G^<(x_1, z_1) G^>(z_1, z_2) G_A(x_2, z_2) G_A(x_4, z_1) G_A(z_2, z_3) G_A(z_3, z_3) G_K(z_2, z_3) G_R(x_3, z_1) \\ \mathcal{T}_{35,2} &= 128ig_2^3 N_c^{-2} G^<(x_1, z_1) G^>(z_1, z_2) G_A(x_2, z_2) G_A(x_4, z_1) G_A(z_2, z_3) G_K(z_2, z_3) G_R(x_3, z_1) G_R(z_3, z_3) \\ \mathcal{T}_{35,3} &= 128ig_2^3 N_c^{-2} G^<(x_1, z_1) G^>(z_1, z_2) G_A(x_2, z_2) G_A(x_4, z_1) G_A(z_2, z_3) G_K(z_3, z_3) G_R(x_3, z_1) G_R(z_2, z_3)\end{aligned}$$

Topology 36 (FeynArts Topology 4)

$2 \rightarrow 2$

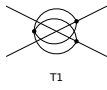


Parsed propagators: 8. Internal vertices: 3. Internal half-edges: 12.
 Assignments checked: 32768. Surviving (nonzero vertex weight): 4096.
 Raw N_c power: 1. Net power in $C(t)$: -3.

No surviving SK contributions for this topology.

Topology 37 (FeynArts Topology 4)

$2 \rightarrow 2$

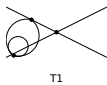


Parsed propagators: 8. Internal vertices: 3. Internal half-edges: 12.
 Assignments checked: 32768. Surviving (nonzero vertex weight): 4096.
 Raw N_c power: 1. Net power in $C(t)$: -3.

No surviving SK contributions for this topology.

Topology 38 (FeynArts Topology 4)

$2 \rightarrow 2$

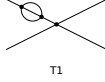


Parsed propagators: 8. Internal vertices: 3. Internal half-edges: 12.
Assignments checked: 32768. Surviving (nonzero vertex weight): 4096.
Raw N_c power: 2. Net power in $C(t)$: -2.

No surviving SK contributions for this topology.

Topology 39 (FeynArts Topology 6)

$2 \rightarrow 2$

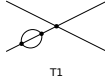


Parsed propagators: 8. Internal vertices: 3. Internal half-edges: 12.
Assignments checked: 32768. Surviving (nonzero vertex weight): 4096.
Raw N_c power: 0. Net power in $C(t)$: -4.

No surviving SK contributions for this topology.

Topology 40 (FeynArts Topology 6)

$2 \rightarrow 2$

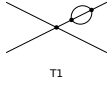


Parsed propagators: 8. Internal vertices: 3. Internal half-edges: 12.
Assignments checked: 32768. Surviving (nonzero vertex weight): 4096.
Raw N_c power: 0. Net power in $C(t)$: -4. Explicit surviving terms:

$$\mathcal{T}_{40,1} = 384ig_2^3 N_c^{-4} G^<(x_1, z_1) G^<(z_3, z_1) G_A(x_2, z_2) G_A(x_4, z_1) G_A(z_3, z_2) G_K(z_3, z_2) G_R(x_3, z_1) G_R(z_3, z_2)$$

Topology 41 (FeynArts Topology 6)

2 → 2



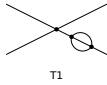
π

Parsed propagators: 8. Internal vertices: 3. Internal half-edges: 12.
 Assignments checked: 32768. Surviving (nonzero vertex weight): 4096.
 Raw N_c power: 0. Net power in $C(t)$: -4.

No surviving SK contributions for this topology.

Topology 42 (FeynArts Topology 6)

2 → 2



π

Parsed propagators: 8. Internal vertices: 3. Internal half-edges: 12.
 Assignments checked: 32768. Surviving (nonzero vertex weight): 4096.
 Raw N_c power: 0. Net power in $C(t)$: -4. Explicit surviving terms:

$$\mathcal{T}_{42,1} = 384ig_2^3 N_c^{-4} G^>(x_3, z_1) G^>(z_3, z_1) G_A(x_2, z_1) G_A(x_4, z_2) G_A(z_3, z_2) G_K(z_3, z_2) G_R(x_1, z_1) G_R(z_3, z_2)$$